

SCHRÖDINGERIZATION BASED COMPUTATIONALLY STABLE ALGORITHMS FOR ILL-POSED PROBLEMS IN PARTIAL DIFFERENTIAL EQUATIONS*

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Abstract. We introduce a simple and stable computational method for ill-posed partial differential equation (PDE) problems. The method is based on Schrödingerization, introduced in [S. Jin, N. Liu, and Y. Yu, *Phys. Rev. Lett.*, 133 (2024), 230602; S. Jin, N. Liu, and Y. Yu, *Phys. Rev. A*, 108 (2023), 032603], which maps all linear PDEs into Schrödinger-type equations in one higher dimension, for quantum simulations of these PDEs. Although the original problem is ill-posed, the Schrödingerized equations are Hamiltonian systems and time-reversible, allowing stable computation both forward and backward in time. The original variable can be recovered by data from suitably chosen domain in the extended dimension. We will use the (constant and variable coefficient) backward heat equation and the linear convection equation with imaginary wave speed as examples. Error analysis of these algorithms are conducted and verified numerically. The methods are applicable to both classical and quantum computers, and we also lay out quantum algorithms for these methods. Moreover, we introduce a smooth initialization for the Schrödingerized equation, which will lead to essentially spectral accuracy for the approximation in the extended space, if a spectral method is used. Consequently, the extra qubits needed due to the extra dimension, if a qubit-based quantum algorithm is used, for both well-posed and ill-posed problems, becomes almost $\log \log 1/\varepsilon$ where ε is the desired precision. This optimizes the complexity of the Schrödingerization based quantum algorithms for any nonunitary dynamical system introduced in [S. Jin, N. Liu, and Y. Yu, *Phys. Rev. Lett.*, 133 (2024), 230602; S. Jin, N. Liu, and Y. Yu, *Phys. Rev. A*, 108 (2023), 032603].

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1. Introduction. In this paper, we are interested in numerically computing ill-posed problems that follow the evolution of a general dynamical system

$$(1.1) \quad \frac{d}{dt} \mathbf{u} = H \mathbf{u}$$

whose input data given by $\mathbf{u}_0 \in \mathbb{R}^n$. For simplicity, we first consider H being Hermitian with n real eigenvalues, ordered as

$$(1.2) \quad \lambda_1(H) \leq \lambda_2(H) \leq \cdots \leq \lambda_n(H) \quad \text{for all } t \in [0, T].$$

We assume that $\lambda_i(H) > 0$, for some $1 \leq i \leq n$, which implies that the system (1.1) contains unstable modes; thus the initial-value problem is ill-posed since its solution will grow exponentially in time. This causes significant computational challenges. Normally the numerical errors will grow exponentially unless special care is taken.

Ill-posed or unstable problems appear in many physical applications, for example, fluid dynamics instabilities such as Rayleigh–Taylor and Kelvin–Helmholtz instabilities [21, 10], plasma instability [12], and Maxwell equations with negative index of refraction [30, 28]. It also appears in inverse problems [33, 4]. One of the most classical ill-posed problems is the backward heat equation which suffers from the catastrophic Hadamard instability. Usually, some regularization technique is used to make the problems well-posed [8].

In this paper, we study a generic yet simple computational strategy to numerically compute ill-posed problems based on Schrödingerization, introduced for quantum simulation for dynamical systems whose evolution operator is nonunitary [16, 17]. The idea is to map it to one higher dimension, into Schrödinger-type equations, that obeys unitary dynamics and is thereby naturally fitting for quantum simulation. Since the Schrödinger-type equations are Hamiltonian systems that are time-reversible, they can be solved both forward and backward in time in a computationally stable way. This makes it suitable for solving unstable problems, as was proposed in our previous work [15]. In this article, we study this method for the classical backward heat equation and a linear convection equation with imaginary wave speed (or negative index of refraction).

The initial-value problem to the backward heat equation is ill-posed in all three ways: (i) the solution does not necessarily exist; (ii) if the solution exists, it is not necessarily unique; (iii) there is no continuous dependence of the solution on arbitrary input data [27, 18, 13, 23]. This problem is well-posed for final data whose Fourier spectrum has compact support [26]. However, even when the solution exists and is unique, computing the solution is difficult since unstable physical systems usually lead to unstable numerical methods. There have been various treatments of this difficult problem, for example, quasi-reversibility methods [34, 31, 22, 3, 25], regularization methods [11, 19, 31, 6], Fourier truncation methods [29], etc. Our approach computes solution consistent with the Fourier truncation method.

The actual implementation of our backward heat equation solver is as follows. First we start with the Fourier transform of the input data denoted by u_T and truncate the Fourier mode in finite domain. This either is achieved automatically if the Fourier mode of u_T has compact support, or we choose a sufficiently large domain in the Fourier space such that outside it the Fourier coefficient of u_T is sufficiently small. The latter can usually be done since the forward heat equation gives a solution that is smooth for $t > 0$ so its Fourier coefficient decays rapidly. The Schrödingerization technique lifts the backward heat equation to a Schrödinger equation in one higher

dimension that is *time-reversible*, which can be solved backward in time by any reasonable stable numerical approximation. We then recover $u(t)$ for $0 \leq t < T$ by integrating or pointwise evaluation over a suitably chosen domain in the extended variable. Since the time evolution is based on solving the Schrödinger equation, the computational method is stable.

We point out that the truncation in the Fourier space regularizes the original ill-posed problem. Although the initial-value problem to the Schrödingerized equation is well-posed even without this Fourier truncation, to recover u one needs *finite* Fourier mode. We also show that the strategy also applies to a variable coefficient heat equation.

We will also apply the same strategy to solve the unstable linear convection equation that has imaginary wave speed.

We will give error estimates on these methods and conduct numerical methods that verify the results of the error analysis. The methods work for both classical and quantum computers. Since the original Schrödingerization method was introduced for quantum simulation of general PDEs, we will also give the quantum implementation of the computational method.

Moreover, we introduce a smooth initialization for the Schrödingerized equation, so the initial data will be in $C^k(\mathbb{R})$ in the extended space for any integer $k \geq 1$ (see subsection 4.3). This will lead to essentially the spectral accuracy for the approximation in the extended space, if a spectral method is used. Consequently, the extra qubits needed due to the extra dimension, if a qubit based quantum algorithm is used, for both well-posed and ill-posed problems, become almost $\log \log 1/\varepsilon$, where ε is the desired precision. This *reduces* the complexity of Schrödingerization-based quantum algorithms introduced in [16, 17] for any nonunitary dynamical system.

The rest of the paper is organized as follows. In section 2, we give a brief review of the Schrödingerization approach for general linear ODEs. In section 3, we study the backward heat equation, for both constant and variable coefficient cases, and show the approximate solution based on the framework of Schrödingerization. In section 4, the numerical method and error analysis of the backward heat equation are presented. In addition, a C^k smooth initial data with respect to the extended variable is constructed to improve the convergence rates in the extended space. In section 5, we apply this technique to the convection equations with purely imaginary wave speed. In section 6, we show the numerical tests to verify our theories. Section 7 shows the quantum algorithms and the corresponding complexity.

Throughout the paper, we restrict the simulation to a finite time interval $t \in [0, T]$. The notation $f \lesssim g$ stands for $f \leq Cg$, where C is independent of the mesh size and time step. Scalar-valued quantities and vector-valued quantities are denoted by normal symbols and boldface symbols, respectively. Moreover, we use a zero-based indexing, i.e., $j = \{0, 1, \dots, N-1\}$, or $j \in [N]$, and $\mathbf{e}_j^{(N)} \in \mathbb{R}^N$ denotes a vector with the j th component being 1 and others 0. We shall denote the identity matrix and null matrix by I and $\mathbf{0}$, respectively, and the dimensions of these matrices should be clear from the context; otherwise the notation I_N stands for the N -dimensional identity matrix.

2. The general framework. We start with the basic framework set up in [15] for forward problems, which we first briefly review here.

Using the warped phase transformation $\mathbf{w}(t, p) = e^{-p} \mathbf{u}$ for $p > 0$ and symmetrically extending the initial data to $p < 0$, one has the following system of linear convection equations [16, 17]:

$$(2.1) \quad \frac{d}{dt} \mathbf{w} = -H \partial_p \mathbf{w} \quad \mathbf{w}(0, p) = e^{-|p|} \mathbf{u}_0.$$

This is clearly a hyperbolic system. When the eigenvalues of H are all negative, the convection term of (2.1) corresponds to waves moving from the right to the left. One does, however, need a boundary condition on the right-hand side. Since $\mathbf{w}$ decays exponentially in p , one just needs to select $p = p^R$ large enough, so w at this point is essentially zero, and a zero incoming boundary condition can be used at $p = p^R$. Then $\mathbf{u}$ can be numerically recovered via

$$(2.2) \quad \mathbf{u}(t) = \frac{1}{1 - e^{-p^R}} \int_0^{p^R} \mathbf{w}(t, p) dp,$$

or

$$(2.3) \quad \mathbf{u}(t) = e^p \mathbf{w}(t, p) \quad \text{for any } 0 < p < p^R.$$

However, when $\lambda_i(H) > 0$ for some i , some of the waves evolving through (2.1) will instead propagate from left to right. If we were solving the problem in domain $p \in [0, \infty)$, we would need a boundary condition at $p = 0$, which is not possible. Instead, we extend the computational domain to also include $[-p^L, 0)$ with a zero incoming boundary at $p = -p^L$ for $p^L > 0$ sufficiently large (so $e^{-p^L} \approx 0$). In summary, we use zero boundary condition for (2.1) restricted on the finite domain $[-p^L, p^R]$. It is worth noting that extending to the $p < 0$ domain is done for the convenience of employing the Fourier spectral method, which requires periodic boundary conditions, and our zero boundary conditions can be regarded, approximately, as periodic boundary conditions. It is important to note that any waves—corresponding to positive eigenvalues of H —that start from $p < 0$ and travel to the right will induce spurious solutions to the domain $p > 0$; therefore, when recovering u , one needs to select the correct domain to avoid using these spurious waves. Therefore, the following theorem [15] must be utilized.

THEOREM 2.1. *Assume the eigenvalues of H satisfy (1.2); then the solution of (1.1) can be recovered by*

$$(2.4) \quad \mathbf{u}(t) = e^p \mathbf{w}(t, p) \quad \text{for any } p > p^\diamond,$$

where $p^\diamond = \max\{\lambda_n(H)t, 0\}$, or recovered by using the integration

$$(2.5) \quad \mathbf{u}(t) = e^p \int_p^\infty \mathbf{w}(t, q) dq, \quad p > p^\diamond.$$

Since (2.1) is a *hyperbolic* system, the initial-value problem is thus *well-posed* and one can solve it numerically in a stable way (as long as suitable numerical stability condition—the CFL condition—is satisfied). Thus although the original system (1.1) is ill-posed, we can still solve the well-posed problem (2.1), and then recover $\mathbf{u}$ using Theorem 2.1, as long as $\lambda_n(H) < \infty$!

We also point out here that the discretization of the p -derivative in (2.1) can be achieved not only by using Fourier spectral methods but also through other numerical schemes, such as finite difference or finite element methods. In such cases one can just solve the problem in $p > 0$, and, if $\lambda_i(H) > 0$ for some i , one needs to supply boundary condition at $p = 0$ for the characteristic fields corresponding to positive eigenvalues of H . This boundary data can be set arbitrarily, since the induced spurious wave propagating with speed $\lambda_i(H) > 0$ will not be used when we recover u using data from $p > p^\diamond$, as laid out by Theorem 2.1.

3. The backward heat equation. In this section, we consider the following one-dimensional backward heat equation in the infinite domain:

$$(3.1) \quad \begin{aligned} \partial_t u &= \partial_{xx} u & \text{in } \Omega_x, \quad T > t \geq 0, \\ u(T, x) &= u_T(x) & \text{in } \Omega_x, \end{aligned}$$

where $\Omega_x = (-\infty, \infty)$. We want to determine $u(t, \cdot)$ for $T > t \geq 0$ from the data u_T . The change of variable $v(t, x) = u(T - t, x)$ leads to the following formulation of (3.1):

$$(3.2) \quad \begin{aligned} \partial_t v &= -\partial_{xx} v & \text{in } \Omega_x, \quad 0 \leq t < T, \\ v(0, x) &= u_T(x) & \text{in } \Omega_x. \end{aligned}$$

It is easy to see that the eigenvalues of H in (1.1) are positive after spatial discretization. Let $\hat{g}(\eta)$ denote the Fourier transform of $g(x) \in L^2(\Omega_x)$ and define it by

$$\hat{g}(\eta) := \mathcal{F}(g) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{ix\eta} g(x) dx, \quad g(x) := \mathcal{F}^{-1}(\hat{g}) = \int_{\mathbb{R}} e^{-ix\eta} \hat{g}(\eta) d\eta.$$

Let $\|g\|_{H^s(\Omega_x)}$ denote the Sobolev norm $H^s(\Omega_x)$ defined as

$$\|g\|_{H^s(\Omega_x)} := \left(\int_{\mathbb{R}} |\hat{g}(\eta)|^2 (1 + \eta^2)^s d\eta \right)^{\frac{1}{2}}.$$

When $s = 0$, $\|\cdot\|_{H^0(\Omega_x)}$ denotes the $L^2(\Omega_x)$ -norm, and s can be noninteger [9]. For the above solution to be well-defined, we make the following assumptions of u_T for analysis:

- (H1) Assume $u_T \in H_0^s$, namely, H^s with compact support. This yields a well-posed problem in a subspace of L^2 in the sense of Hadamard [26].
- (H2) For more general $\hat{u}_T$, we will truncate the domain in η and begin with the truncated—thus compactly supported—final data.

We remark here that truncation is inherently achieved in our methodology (see (S1) and (S2)), thus providing a form of *regularization* to the original ill-posed problem.

If the solution of (3.1) in this Sobolev space exists, then it must be unique [9]. We assume $u(t, x)$ is the unique solution of (3.1). Applying the Fourier transform, one gets the exact solution $u(t, x)$ of problem (3.1):

$$(3.3) \quad u(t, x) = \int_{\mathbb{R}} e^{-ix\eta} e^{\eta^2(T-t)} \hat{u}_T d\eta.$$

Denoting $u_0(x) = u(0, x)$, then it is easy to see from (3.3) that

$$(3.4) \quad \|u_0\|_{H^s(\Omega_x)} = \left(\int_{\mathbb{R}} |\hat{u}_0|^2 (1 + \eta^2)^s d\eta \right)^{1/2} < \infty \quad \text{for } s \geq 1.$$

The Schrödingerization for (3.1) gives

$$(3.5) \quad \begin{aligned} \frac{d}{dt} w &= -\partial_{xp} w & \text{in } \Omega_x \times \Omega_p, \quad T > t \geq 0, \\ w(T, x, p) &= e^{-|p|} u_T(x) & \text{in } \Omega_x \times \Omega_p, \end{aligned}$$

where $\Omega_p = (-\infty, \infty)$. Again using the Fourier transform technique to (3.5) with respect to the variables p and x , one gets the Fourier transform $\hat{w}(t, \eta, \xi)$ of the exact solution $w(t, x, p)$ of problem (3.5) satisfying

$$(3.6) \quad \frac{d}{dt} \hat{w} = -i\xi\eta^2 \hat{w}, \quad \hat{w}(T, \eta, \xi) = \hat{w}_T = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{e^{ix\eta} u_T(x)}{\pi(1 + \xi^2)} dx = \frac{\hat{u}_T(\eta)}{\pi(1 + \xi^2)}.$$

The solution of w is then found to be

$$(3.7) \quad w(t, x, p) = \int_{\mathbb{R}} \int_{\mathbb{R}} e^{-i(x\eta + p\xi)} e^{-i\xi\eta^2(t-T)} \hat{w}_T \, d\eta d\xi.$$

Note the initial-value problem (3.5) is well-posed, *even for $u_T \in H^s$ not being compactly supported in the Fourier space*, as can be easily seen from (3.6), which is an oscillatory ODE with purely imaginary spectra. Following the proof in [9, Theorem 5, section 7.3], one has the regularity of $w(0, x, p) \in H^s(\Omega_x) \times H^1(\Omega_p)$ under the assumption in (H1) or (H2).

Using the Fourier transform only on x gives

$$(3.8) \quad \partial_t \hat{w}^x - \eta^2 \partial_p \hat{w}^x = 0, \quad T > t \geq 0, \quad \hat{w}^x(T, \eta, p) = e^{-|p|} \hat{u}_T(\eta),$$

where $\hat{w}^x$ is the Fourier transform of w on the x -variable, i.e.,

$$(3.9) \quad \hat{w}^x(t, \eta, p) = \frac{1}{2\pi} \int_{\mathbb{R}} w(t, x, p) e^{ix\eta} dx.$$

Here $\hat{w}^x$ is a linear wave moving to the right, if one starts from $t = T$ and going backward in time (one can also use the change of variable $t \mapsto T - t$ to make the problem (3.5) forward in time for notation comfort), and the solution is computed by the inverse Fourier transformation

$$(3.10) \quad w(t, x, p) = \int_{\mathbb{R}} e^{-|p + \eta^2(t-T)|} \hat{u}_T(\eta) e^{-ix\eta} d\eta.$$

This corresponds to the case of (2.1) in which all eigenvalues of H are *positive*. Therefore, to recover $u(0, x)$ from (3.10), one needs to choose $p > \eta^2 T$ for time $T > 0$ to obtain

$$u(0, x) = e^p \int_{\mathbb{R}} e^{-|p - \eta^2 T|} \hat{u}_T e^{-ix\eta} d\eta = \int_{\mathbb{R}} e^{\eta^2 T} \hat{u}_T e^{-ix\eta} d\eta, \quad p > \eta^2 T.$$

This is the basis for the introduction of our stable computational method: we solve the w -equation (3.5) with a suitable—stable—computational method, and then recover u by either integration or pointwise evaluation using Theorem 2.1. However, in the continuous space η maybe unbounded, and thus the condition $p > \eta^2 T$ may not be satisfied. We consider two scenarios here:

- (S1) The input data $\hat{u}_T$ has compact support, namely, $\hat{u}_T = 0$ for $|\eta| > \eta_0$ for some $\eta_0 < \infty$. While this is not generally true, when $u_T(x)$ is smooth in x , its Fourier mode decays rapidly, so one can truncate η for $|\eta| > \eta_0$ with desired accuracy.
- (S2) One discretizes (3.5) numerically. This usually requires one first to truncate the x -domain so it is finite, followed by some numerical discretization of the x -derivative, for example, the finite difference or spectral method. Then $|\eta| \leq \eta_0 \leq \mathcal{O}(1/(\Delta x)^2)$, where Δx is the spatial mesh size in x .

3.1. Error estimates. We start by choosing an $\eta_{\max} > 0$ and denote

$$(3.11) \quad \delta(\eta_{\max}) = \left(\int_{\eta_{\max}}^{\infty} (1 + \eta^2)^s |\hat{u}_0|^2 d\eta \right)^{\frac{1}{2}}$$

to be a small quantity, which will be chosen with other error terms to meet the numerical tolerance requirement (see Remark 3.4). We define the “approximate” solution $u_{\eta_{\max}}$ of (3.1) as follows.

DEFINITION 3.1. *First, define*

$$(3.12) \quad w_{\eta_{\max}} = \int_{\mathbb{R}} e^{-|p+\eta^2(t-T)|} \hat{u}_T(\eta) \chi_{\max} e^{-ix\eta} d\eta,$$

where $\chi_{\max}(\eta) = 1$ when $\eta \in [-\eta_{\max}, \eta_{\max}]$; otherwise $\chi(\eta) = 0$. Then one obtains the approximate solution $u_{\eta_{\max}}$ by

$$(3.13) \quad u_{\eta_{\max}}(t, x, p) = e^p w_{\eta_{\max}}(t, x, p), \quad \text{or} \quad u_{\eta_{\max}}(t, x, p) = e^p \int_p^\infty w(t, x, q) dq,$$

with $p \geq p^\diamond = \eta_{\max}^2 T$.

With u defined as above, we have the following theorem which gives an error estimate to the above approximation.

THEOREM 3.2. *Suppose $u_T \in H^s(\Omega_x)$ satisfies the assumption in (H1) or (H2). Let $u(t) = u(t, x)$ be the solution (3.1) given by (3.3), and let $u_{\eta_{\max}}(t) = u_{\eta_{\max}}(t, x, p)$ be given by (3.13) with $p > p^\diamond = \eta_{\max}^2 T$ and $\eta_{\max}$ chosen to satisfy (3.11). Then for any $p > p^\diamond$, $t \in [0, T]$,*

$$(3.14) \quad \|u(t) - u_{\eta_{\max}}(t)\|_{L^2(\Omega_x)} \leq e^{-\eta_{\max}^2 t} \delta(\eta_{\max}) / \eta_{\max}^s.$$

Proof. If u_T satisfies (H1) or (H2), it is apparent that $\delta(\eta_{\max})$ for some $\eta_{\max}$ in (3.11) is well-defined. According to Parseval's equality and (3.3), (3.11)–(3.13), it follows that

$$\begin{aligned} \|u - u_{\eta_{\max}}\|_{L^2(\Omega_x)} &= \|\hat{u} - \hat{u}_{\eta_{\max}}\|_{L^2(\Omega_\eta)} = \|e^{\eta^2(T-t)} \hat{u}_T - e^{\eta^2(T-t)} \hat{u}_T \chi_{\max}\|_{L^2(\Omega_\eta)} \\ &= \left(\int_{|\eta| > \eta_{\max}} |e^{\eta^2(T-t)} e^{-\eta^2 T} \hat{u}_0|^2 dx \right)^{\frac{1}{2}} \\ &= \left(\int_{|\eta| > \eta_{\max}} |e^{-\eta^2 t} \hat{u}_0|^2 dx \right)^{\frac{1}{2}} \\ &= \left(\int_{|\eta| > \eta_{\max}} \frac{e^{-\eta^2 t} |\hat{u}_0|^2}{(1 + \eta^2)^s} (1 + \eta^2)^s d\eta \right)^{\frac{1}{2}} \\ (3.15) \quad &\leq \left(\int_{|\eta| > \eta_{\max}} \frac{e^{-\eta_{\max}^2 t} |\hat{u}_0|^2}{\eta^{2s}} (1 + \eta^2)^s d\eta \right)^{\frac{1}{2}} \leq \frac{e^{-\eta_{\max}^2 t} \delta(\eta_{\max})}{\eta_{\max}^s}. \end{aligned}$$

The proof is completed. $\square$

Remark 3.3. Since we are solving the problem (3.1) *backward in time*, the error estimate obtained in Theorem 3.2 needs to be understood also backward in time, namely the error will *increase* as t goes from T to 0. Numerical experiments in subsection 6.2 also confirms this.

Remark 3.4. In order to bound the error by ε at $t = 0$, the shape estimate of $\eta_{\max}$ from (3.15) is that $\eta_{\max}$ is chosen such that

$$(3.16) \quad \left(\int_{|\eta| > \eta_{\max}} |\hat{u}_0|^2 dx \right)^{\frac{1}{2}} \leq \varepsilon.$$

If $u_0 \in H^s(\Omega_x)$, one could choose $\eta_{\max}$ satisfying

$$(3.17) \quad \eta_{\max} \geq \left(\frac{\delta}{\varepsilon}\right)^{\frac{1}{s}}.$$

Considering $u_0 \in H^s(\Omega_x)$, one may have $|\hat{u}_0|^2(1+\eta^2)^s = \mathcal{O}(\frac{1}{\eta^{1+\epsilon_0}})$ with $0 < \epsilon_0 \leq 2$, and then $\delta(\eta_{\max}) = \mathcal{O}(1/(\eta_{\max}^{\epsilon_0/2}))$ is small. Besides, if u_0 is smooth enough, $p^\diamond = (\frac{\delta}{\varepsilon})^{\frac{2}{s}}T$ is not necessarily large.

Specifically, if the Fourier transform of u has compact support such that $\hat{u}_T(\eta) = 0$ when $\eta > \eta_0$, it yields from (3.15)–(3.16) that there exists no error of recovery by choosing $p^\diamond = \eta_0^2 T$. To take an example, $u = \alpha(t) \cos(\omega_0 x)$ (or $u = \alpha(t) \sin(\omega_0 x)$) is a periodic function in $[-\frac{\pi}{\omega_0}, \frac{\pi}{\omega_0}]$. We extend it periodically over the whole field of real domain $\mathbb{R}$. The Fourier transform of u is $\hat{u}(t, \eta) = \alpha(t) \pi(\delta(\eta - \omega_0) + \delta(\eta + \omega_0))$, where $\delta(\eta + \omega_0)$ gives an impulse that is shifted to the left by ω_0 ; likewise the function $\delta(\eta - \omega_0)$ yields an impulse shifted to the right by ω_0 . In this case, $\hat{u}(t, \eta) \equiv 0$ when $|\eta| > \omega_0 = \eta_0$. Therefore, the best choice of $p^\diamond$ for $u = \alpha(t) \cos(\omega_0 x)$ (or $u = \alpha(t) \sin(\omega_0 x)$) is

$$(3.18) \quad p^\diamond = \omega_0^2 T = \eta_0^2 T.$$

At this point, $\eta_{\max} = \omega_0 = \eta_0$, one has $\delta(\eta_{\max}) = 0$, $u = u_{\eta_{\max}}$ for $p > p^\diamond$.

In practice, the data at $t = T$ is usually imprecise since it depends on the reading of physical measurements. Consider a perturbed data u_T^ζ , which is a small disturb of u_T . The Fourier transform of u_T^ζ may not decay as $|\eta| \rightarrow \infty$, which leads to the severely ill-posed problems of (3.1). However, after Schrödingerization, a disturb of the Fourier transform of w defined by $\hat{w}_T^\zeta = \frac{\hat{u}_T^\zeta(\eta)}{\pi(1+\xi^2)}$ does not affect the well-posedness of w even if w_T^ζ is merely in $L^2(\mathbb{R})$. Assume the measured data u_T^ζ satisfies

$$(3.19) \quad \|u_T^\zeta - u_T\|_{L^2(\Omega_x)} \leq \zeta_0.$$

Define the approximate solution at time t from u_T^ζ by

$$(3.20) \quad \begin{aligned} w_{\eta_{\max}, \zeta} &= \int_{\mathbb{R}} \int_{\mathbb{R}} e^{-i(x\eta + p\xi)} e^{-i\xi\eta^2(t-T)} \hat{w}_T^\zeta \chi_{\max} \, d\eta d\xi \\ &= \int_{\mathbb{R}} e^{-|p+\eta^2(t-T)|} \hat{u}_T^\zeta(\eta) e^{-ix\eta} \chi_{\max} \, d\eta. \end{aligned}$$

Now we have the approximate solution $w_{\eta_{\max}, \zeta}$.

THEOREM 3.5. *Suppose $u_T \in H^s(\Omega_x)$ satisfies the assumption in (H1) or (H2). Let $u(t, x)$ be given by (3.1) and $w_{\eta_{\max}, \zeta}(t, x, p)$ be given by (3.20), respectively. If $u_{\eta_{\max}, \zeta}$ is recovered by*

$$(3.21) \quad u_{\eta_{\max}, \zeta} = e^p w_{\eta_{\max}, \zeta}(t, x, p), \quad \text{or} \quad u_{\eta_{\max}, \zeta} = e^p \int_p^\infty w_{\eta_{\max}, \zeta}(t, x, q) \, dq,$$

for any $p > p^\diamond = \eta_{\max}^2 T$, and $\eta_{\max}$ is chosen such that $\frac{\delta(\eta_{\max})}{|\eta_{\max}|^s} + e^{\eta_{\max}^2 T} \zeta_0 \leq \varepsilon$, then

$$\|u(t) - u_{\eta_{\max}, \zeta}(t)\|_{L^2(\Omega_x)} \leq \varepsilon.$$

Proof. From the definition of $u_{\eta_{\max}, \zeta}$ in (3.21), one has

$$u_{\eta_{\max}, \zeta} = \int_{\mathbb{R}} e^{-ix\eta} e^{\eta^2(T-t)} \hat{u}_T^\zeta(\eta) \chi_{\max} d\eta,$$

for $p > p^\diamond$. Then one has

(3.22)

$$\|u_{\eta_{\max}, \zeta} - u_{\eta, \max}\|_{L^2(\Omega_x)} = \left(\int_{|\eta| \leq \eta_{\max}} |e^{\eta^2(T-t)} (\hat{u}_T^\zeta - \hat{u}_T)|^2 d\eta \right)^{\frac{1}{2}} \leq e^{\eta_{\max}^2(T-t)} \zeta_0.$$

The proof is finished by combining (3.15) and (3.22). $\square$

3.2. Variable coefficient problems. When the system has a spatially varying coefficient, the Fourier transform cannot be applied, and the analysis in Theorem 3.2 no longer holds. However, the Schrödingerization still works. We consider variable coefficient equation

$$(3.23) \quad \partial_t u = \partial_x(a(x)\partial_x u) \quad x \in \Omega_x, \quad 0 < t < T, \quad u(T, x) = u_T(x),$$

with Dirichlet's boundary condition of bounded domain Ω_x in straightforward way, where $a(x) \in C^\infty(\bar{\Omega}_x)$ and $a(x) \geq \alpha_0 > 0$.

Let ϕ_k be the k th eigenfunction corresponding to λ_k of the operator $\mathcal{L} := -\partial_x(a(x)\partial_x \cdot)$ such that

$$(3.24) \quad \mathcal{L}\phi_k = \lambda_k \phi_k, \quad \phi_k|_{\partial\Omega_x} = 0.$$

It is easy to see that $\phi_k \in C^\infty(\Omega_x)$ from the regularity of the second-order elliptic equations [9, Theorem 3, section 6.3]. They form an orthonormal basis set of $L^2(\Omega_x)$. According to standard theory on the eigenvalues of symmetric elliptic operators [9, Theorem 1, section 6.5], one has

$$(3.25) \quad 0 < \lambda_1 \leq \lambda_2 \leq \lambda_3 \leq \cdots, \quad \lambda_k \rightarrow \infty \quad \text{as } k \rightarrow \infty.$$

We search for solution to (3.23) at time t from the data u_T in the form

$$(3.26) \quad u(t, x) = \sum_{k \in \mathbb{N}^+} \alpha_k(t) \phi_k(x) = \sum_{k \in \mathbb{N}^+} e^{\lambda_k(T-t)} \alpha_k(T) \phi_k(x) \quad T > t \geq 0,$$

where $\alpha_k(T) = \int_{\Omega_x} u_T \phi_k dx$.

Define the equivalent norm of $\|\cdot\|_{H^s}$, $s \in \mathbb{N}$, again denoted by $\|\cdot\|_{H^s}$:

$$(3.27) \quad \|v\|_{H^s(\Omega_x)}^2 = \sum_{k \in \mathbb{N}^+} |\beta_k|^2 (1 + \lambda_k + \cdots \lambda_k^s),$$

where $\beta_k = \int_{\Omega_x} v \phi_k dx$. Here $\|\cdot\|_{H^0(\Omega_x)} := \|\cdot\|_{L^2(\Omega_x)}$ denotes $L^2(\Omega_x)$ norm when $s = 0$, and s can also be a noninteger for $1 \leq s \leq 2$ [20, Proposition 2.1]. For the solution to (3.23) to be well-defined, we assume the expansion of $u_T \in H^s(\Omega_x)$ to the orthonormal basis has only finite number $n_{\max}$ of terms, leading to $u(t, x) \in H^s(\Omega_x)$ for all $0 \leq t \leq T$; i.e.,

$$(3.28) \quad \begin{aligned} \|u(t, \cdot)\|_{H^s(\Omega_x)}^2 &= \sum_{k \leq n_{\max}} |\alpha_k(t)|^2 (1 + \lambda_k + \cdots \lambda_k^s) \\ &= \sum_{k \leq n_{\max}} |e^{\lambda_k(T-t)} \alpha_k(T)|^2 (1 + \lambda_k + \cdots \lambda_k^s) < \infty. \end{aligned}$$

For more general cases, we truncate to the finite terms of expansions of the input data. This regularizes the originally ill-posed problem. Next, assume $w(t, x, p)$ is in the form of

$$w(t, x, p) = \sum_{k \in \mathbb{N}^+} w_k(t, p) \phi_k(x).$$

We correspondingly project the input data $w(T, x, p)$ on the same basis set, and (3.5) is equivalent to

$$\frac{d}{dt} w_k(t, p) = -\lambda_k \partial_p w_k(t, p), \quad w_k(T, p) = e^{-|p|} \alpha_k(T).$$

Note w is well-defined even if the series is not truncated. We then define the truncated approximate solution by

$$(3.29) \quad w_{n_{\max}} = \sum_{1 \leq k \leq n_{\max}} e^{-|p - \lambda_k(T-t)|} \alpha_k(T) \phi_k(x).$$

Letting

$$(3.30) \quad \delta(n_{\max}) = \left(\sum_{k=n_{\max}}^{\infty} |\alpha_k(0)|^2 \lambda_k^s \right)^{1/2},$$

we choose $n_{\max}$ such that

$$(3.31) \quad \delta(n_{\max}) / \lambda_{n_{\max}}^{\frac{s}{2}} \leq \varepsilon$$

for the desired precision ε . We remark here $n_{\max}$ exists from (3.28) if $u_T \in H^s(\Omega_x)$.

THEOREM 3.6. Assume $u_T \in H^s(\Omega_x)$ with only finite number terms of expansion to the orthonormal basis, or if not, truncated to be so. Let $u(t, \cdot)$ be the solution to (3.23) given by (3.26) and $w_{n_{\max}}(t, x, p)$ be given by (3.29), respectively. If $u_{n_{\max}}$ is recovered by

$$(3.32) \quad u_{n_{\max}} = e^p w_{n_{\max}} \quad \text{or} \quad u_{n_{\max}} = e^p \int_p^\infty w_{n_{\max}}(t, x, q) dq,$$

where $p > p^\diamond = \lambda_{n_{\max}} T$, and $n_{\max}$ is chosen by (3.31), then

$$(3.33) \quad \|u(t, \cdot) - u_{n_{\max}}(t, \cdot)\|_{L^2(\Omega_x)} \leq e^{-\lambda_{n_{\max}} t} \delta(n_{\max}) / \lambda_{n_{\max}}^{\frac{s}{2}} \leq \varepsilon.$$

The proof is similar to Theorem 3.2 by using the spectral theory of the symmetric elliptic operators and the norm defined in (3.27). We omit it here.

4. Discretization of the Schrödingerized equation. In this section, we consider the discretization of (3.5) and the corresponding error estimates. For simplicity, we consider u to be a periodic function defined in $\Omega_x = [-\pi, \pi]$.

4.1. The numerical discretization. We discretize the x domain uniformly with the mesh size $\Delta x = 2\pi/M$, where M is a positive even integer and the grid points are denoted by $-\pi = x_0 < \dots < x_M = \pi$. The 1-D basis functions for the Fourier spectral method are usually chosen as

$$(4.1) \quad \phi_j(x) = e^{i\mu_j(x+\pi)}, \quad \mu_j = (j - M/2), \quad j \in [M].$$

Considering the Fourier spectral discretization on x , one easily gets

$$(4.2) \quad \frac{d}{dt} \mathbf{u}_h = -\Phi A(\Phi)^{-1} \mathbf{u}_h, \quad \Phi = (\phi_{ij})_{M \times M} = (\phi_j(x_i))_{M \times M},$$

where $\mathbf{u}_h(T) = [u_T(x_0), u_T(x_1), \dots, u_T(x_{M-1})]^\top$ and $A = D_x^2$. The matrix D_x is obtained by $D_x = \text{diag}\{\mu_0, \dots, \mu_{M-1}\}$. By applying matrix exponentials, one has

$$(4.3) \quad \mathbf{u}_h(t) = \Phi e^{A(T-t)} (\Phi)^{-1} \mathbf{u}_h(T).$$

However, it is difficult to obtain the numerical solution $\mathbf{u}_h$ in the classical computer from (4.3) since the problem is unstable.

We now introduce the discretization of p domain as in [15], first truncating the p -region to $[-\pi L, \pi L]$, where πL is large enough such that

$$(4.4) \quad e^{-(\pi L - 4.5\eta_{\max}^2 T)} \|u_T\|_{L^2(\Omega_x)} \leq \mathcal{O}(\varepsilon),$$

with ε error bound. We point out that the constant 4.5 is just for numerical analysis which is quite mild for numerical tests, and it is sufficient to take it as approximately 1. Then, we can treat w at the boundary as $w(t, x, \pi L) \approx w(t, x, -\pi L) \approx 0$. Using the spectral method, one gets the transformation and difference matrix

$$(\Phi_p)_{ij} = (e^{i\mu_i^p(p_j + \pi L)}) \in \mathbb{C}^{N \times N}, \quad D_p = \text{diag}\{\mu_0^p, \mu_1^p, \dots, \mu_{N-1}^p\}, \quad \mu_i^p = (i - N/2)/L,$$

where $p_j = -\pi L + j\Delta p$ with $\Delta p = \frac{2\pi L}{N}$. Applying the discrete Fourier spectral discretization on p and x , i.e., $\Phi_{px} = \Phi^p \otimes \Phi$, it yields

$$(4.5) \quad \frac{d}{dt} \mathbf{w}_h = i\Phi_{px}(D_p \otimes A)\Phi_{px}^{-1} \mathbf{w}_h, \quad \mathbf{w}_h(T) = \mathbf{g}_h \otimes \mathbf{u}_h(T),$$

where $\mathbf{g}_h = [e^{-|p_0|}, e^{-|p_1|}, \dots, e^{-|p_{N-1}|}]^\top$. The matrices Φ^{-1} (Φ) or Φ_p^{-1} (Φ_p) can be implemented by (inverse) fast Fourier transforms. Since $D_p \otimes A$ is a diagonal matrix, the solution at time t is computed by

$$(4.6) \quad \mathbf{w}_h(t) = \Phi_{px} \mathcal{U}(T-t) \Phi_{px}^{-1} \mathbf{w}_h(T), \quad \mathcal{U}(T-t) = \exp(iD_p \otimes A(T-t)).$$

This approach ensures that no error is introduced in the time discretization. The numerical solution to the system (3.1) is

$$(4.7) \quad \mathbf{u}_h = e^{pk_0} \left(\left(\mathbf{e}_{k_0}^{(N)} \right)^\top \otimes I_M \right) \mathbf{w}_h,$$

where $k_0 = \min\{k : p_k > p^\diamond = \eta_{\max}^2 T\}$ from Theorem 3.2, or it can be recovered by second-order numerical integration

$$(4.8) \quad \mathbf{u}_h = \frac{\sum_{j \in \mathcal{J}} \Delta p \left(\left(\mathbf{e}_j^{(N)} \right)^\top \otimes I_M \right) \mathbf{w}_h}{e^{-(p_{j_0} - 0.5\Delta p)} - e^{-(\pi L - 0.5\Delta p)}},$$

with $j_0 = \min \mathcal{J} = \min\{j : p_j - 0.5\Delta p > p^\diamond = \eta_{\max}^2 T\}$.

4.2. Error analysis for the spatial discretization. Following the general error estimates of Schrödingerization in the extended domain in [15], we derive the specific estimates for the above approximation to the backward heat equation under the assumption that $u_T \in H^s(\Omega_x)$ satisfies (H1) or (H2).

Define the complex $(N+1)$ and $(M+1)$ -dimensional space with respect to p and x , respectively:

$$(4.9) \quad X_N^p = \text{span} \left\{ e^{ik(p/L)} : -\frac{N}{2} \leq k \leq \frac{N}{2} \right\}, \quad X_M^x = \text{span} \left\{ e^{ilx} : -\frac{M}{2} \leq l \leq \frac{M}{2} \right\}.$$

The approximation of w in the finite space $X_N^p \times X_M^x$ from the numerical solution in (4.5) is

$$(4.10) \quad w_h^d(t, x, p) = \sum_{|k| \leq \frac{N}{2}} \sum_{|l| \leq \frac{M}{2}} \tilde{w}_{k,l} e^{ik(\frac{p}{L} + \pi)} e^{il(x + \pi)}, \quad \tilde{w}_{k,l} = (e_{j_k}^{(N)} \otimes e_{j_l}^{(M)})^\top \Phi_{px}^{-1} \mathbf{w}_h(t),$$

where $j_k = -k + N/2$, $j_l = -l + M/2$. Correspondingly, the approximation of u is defined by

$$(4.11) \quad u_h^d(t, x, p) = e^p w_h^d(t, x, p), \quad \text{or} \quad u_d^*(t, x, p) = \frac{1}{e^{-p} - e^{-\pi L}} \int_p^{\pi L} w_h^d(t, x, q) dq,$$

where $p > p^\diamond = \eta_{\max}^2 T$.

It is apparent that the discretization serves as an approximation for the following system with periodic boundary conditions:

$$(4.12) \quad \begin{cases} \frac{d}{dt} \mathcal{W} = -\partial_{xpp} \mathcal{W} & \text{in } \Omega_x \times \Omega_p, \quad T > t \geq 0, \\ \mathcal{W}(t, x, -\pi L) = \mathcal{W}(t, x, \pi L), \quad \mathcal{W}(t, -\pi, p) = \mathcal{W}(t, \pi, p), \\ \mathcal{W}(T, x, p) = \mathcal{G}(p) u_T(x), \end{cases}$$

where $\Omega_p = (-\pi L, \pi L)$, and $\mathcal{G}$ is periodic with a period of $2\pi L$, such that

$$(4.13) \quad \mathcal{G}(p) = e^{-|p-2m\pi L|}, \quad (2m-1)\pi L \leq p < (2m+1)\pi L, \quad m \in \mathbb{Z}.$$

Since $\mathcal{W}$ is also periodic with respect to x , the Fourier transform on variable x of $\mathcal{W}$ is defined by

$$(4.14) \quad \hat{\mathcal{W}}^x(t, \eta, p) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{ix\eta} \mathcal{W}(t, x, p) dx.$$

According to the fundamental results on regularity of transport equations, it is well known that the initial-value problem (4.12) is well-posed, and $\mathcal{W}(0, x, p) \in H^s(\Omega_x) \times H^1(\Omega_p)$ satisfies [9, Theorem 5, section 7.3]

$$(4.15) \quad \|\mathcal{W}(0, x, p)\|_{H^s(\Omega_x) \times H^1(\Omega_p)} \lesssim \|\mathcal{G}\|_{H^1(\Omega_p)} \|u_0\|_{H^s(\Omega_x)} \lesssim \|w_0\|_{H^s(\Omega_x) \times H^1(\Omega_p)}.$$

In addition, the standard estimate of spectral methods shows

$$(4.16) \quad \begin{aligned} \|w_h^d - \mathcal{W}(0, x, p)\|_{L^2(\Omega_p) \times L^2(\Omega_x)} &\lesssim (\Delta p + \Delta x^s) \|\mathcal{W}(0, x, p)\|_{H^s(\Omega_x) \times H^1(\Omega_p)} \\ &\lesssim (\Delta p + \Delta x^s) \|w_0\|_{H^s(\Omega_x) \times H^1(\Omega_p)}. \end{aligned}$$

LEMMA 4.1. Suppose $\eta_{\max}$ satisfies (3.17), $\pi L > 3\eta_{\max}^2 T$, and $u_T \in H^s(\Omega_x)$ satisfies (H1) or (H2). Let $\mathcal{E}_w(t, x, p) = \mathcal{W}(t, x, p) - w(t, x, p)$. It follows that

$$\int_0^T |\mathcal{E}_w(t, \cdot, -\pi L)|_{H^1(\Omega_x)}^2 dt \leq \frac{5}{2} e^{-2(\pi L - 3\eta_{\max}^2 T)} \|u_T\|_{L^2(\Omega_x)}^2 + e^{-3\eta_{\max}^2 T} \varepsilon^2.$$

Proof. It is easy to obtain from the system in (3.7) that

$$(4.17) \quad \hat{w}^x(t, \eta, p) = e^{-|p+\eta^2(t-T)|} \hat{u}_T(\eta) = e^{-|p+\eta^2(t-T)|-\eta^2 T} \hat{u}_0(\eta).$$

Similarly, one has

$$(4.18) \quad \hat{\mathcal{W}}^x(t, \eta, p) = \mathcal{G}(p + \eta^2(t - T)) \hat{u}_T(\eta) = \mathcal{G}(p + \eta^2(t - T)) e^{-\eta^2 T} \hat{u}_0(\eta).$$

Using the semi- H^1 norm in phase space, it yields

$$|\mathcal{E}_w(t, \cdot, -\pi L)|_{H^1(\Omega_x)}^2 = \int_{\mathbb{R}} \eta^2 |\hat{w}^x(t, \eta, -\pi L) - \hat{\mathcal{W}}^x(t, \eta, -\pi L)|^2 d\eta.$$

According to (4.17) and (4.18), changing the order of integration and letting $q = -\pi L + \eta^2(t - T)$ yields

$$(4.19) \quad \int_0^T |\mathcal{E}_w(t, \cdot, -\pi L)|_{H^1(\Omega_x)}^2 dt = I_1 + I_2,$$

where I_1 and I_2 are computed by

$$I_1 = \int_{|\eta| \leq \sqrt{3}\eta_{\max}} \hat{u}_T^2(\eta) \int_{-\pi L - \eta^2 T}^{-\pi L} |e^q - \mathcal{G}(q)|^2 dq d\eta,$$

$$I_2 = \int_{|\eta| > \sqrt{3}\eta_{\max}} \hat{u}_T^2(\eta) \int_{-\pi L - \eta^2 T}^{-\pi L} |e^q - \mathcal{G}(q)|^2 dq d\eta.$$

Since $\mathcal{G}(q)$ is periodic with a period of $2\pi L$ defined in (4.13), $\pi L > 3\eta_{\max}^2 T$, and $\eta_{\max}$ satisfies (3.17), one has

$$(4.20) \quad I_1 = \int_{|\eta| \leq \sqrt{3}\eta_{\max}} \hat{u}_T^2 \int_{-\pi L - \eta^2 T}^{-\pi L} |e^q - e^{-(q+2\pi L)}|^2 dq d\eta$$

$$\leq \frac{1}{2} e^{-2(\pi L - 3\eta_{\max}^2 T)} \|u_T\|_{L^2(\Omega_x)}^2,$$

$$I_2 = \int_{|\eta| > \sqrt{3}\eta_{\max}} \hat{u}_T^2 \int_{-\pi L - \eta^2 T}^{-\pi L} |e^q - \mathcal{G}(q)|^2 dq d\eta$$

$$\leq 2 \int_{|\eta| > \sqrt{3}\eta_{\max}} \left(\hat{u}_T^2 \int_{-\pi L - \eta^2 T}^{-\pi L} e^{2q} dq + e^{-2\eta^2 T} \hat{u}_0^2 \int_{-\pi L - \eta^2 T}^{-\pi L} |\mathcal{G}(q)|^2 dq \right) d\eta$$

$$\leq 2e^{-2\pi L} \|u_T\|_{L^2(\Omega_x)}^2 + \int_{|\eta| > \sqrt{3}\eta_{\max}} e^{-\eta^2 T} \hat{u}_0^2 d\eta$$

$$(4.21) \quad \leq 2e^{-2\pi L} \|u_T\|_{L^2(\Omega_x)}^2 + e^{-3\eta_{\max}^2 T} \varepsilon^2.$$

The proof is completed by inserting (4.20) and (4.21) into (4.19). $\square$

LEMMA 4.2. Assume the assumptions in Lemma 4.1 hold. With $w(t, x, p)$ and $\mathcal{W}(t, x, p)$ defined in (3.5) and (4.12), respectively, it follows that

$$(4.22) \quad \|w_0 - \mathcal{W}_0\|_{L^2(\Omega_x) \times L^2(\Omega_p)} \lesssim e^{-(\pi L - 3\eta_{\max}^2 T)} \|u_T\|_{L^2(\Omega_x)} + e^{-3/2\eta_{\max}^2 T} \varepsilon,$$

where $w_0 = w(0, x, p)$ and $\mathcal{W}_0 = \mathcal{W}(0, x, p)$.

Proof. By introducing $\mathcal{E}_w = \mathcal{W} - w$, one has

$$(4.23) \quad \frac{d}{dt} \mathcal{E}_w = -\partial_{x p} \mathcal{E}_w \quad \mathcal{E}_w(T, x, p) = 0 \text{ in } \Omega_x \times \Omega_p.$$

Multiplying (4.23) by $\mathcal{E}_w$, it follows that

$$\int_0^T \int_{\Omega_x \times \Omega_p} \partial_t \mathcal{E}_w \mathcal{E}_w dx dp dt = \int_0^T \int_{\Omega_x \times \Omega_p} (-\partial_{x p} \mathcal{E}_w) \mathcal{E}_w dx dp dt.$$

An integration by parts gives

$$(4.24) \quad \int_0^T \frac{d}{dt} \|\mathcal{E}_w\|_{L^2(\Omega_x) \times L^2(\Omega_p)}^2 dt = \int_0^T |\mathcal{E}_w(t, \cdot, \pi L)|_{H^1(\Omega_x)}^2 - |\mathcal{E}_w(t, \cdot, -\pi L)|_{H^1(\Omega_x)}^2 dt.$$

From Lemma 4.1 and (4.24), one obtains

$$\begin{aligned} \|\mathcal{E}_w(0, \cdot, \cdot)\|_{L^2(\Omega_x) \times L^2(\Omega_p)} &\leq \left(\int_0^T |\mathcal{E}_w(t, \cdot, -\pi L)|_{H^1(\Omega_x)}^2 dt \right)^{1/2} \\ &\lesssim e^{-(\pi L - 3\eta_{\max}^2 T)} \|u_T\|_{L^2(\Omega_x)} + e^{-3/2\eta_{\max}^2 T} \varepsilon. \end{aligned}$$

The proof is finished. $\square$

THEOREM 4.3. Suppose $\eta_{\max}$ satisfies (3.17), πL is large enough to satisfy (4.4), and $u_T \in H^s(\Omega_x)$ satisfies (H1) or (H2). Define $u_h^d = e^p w_h^d$ with $p \in \tilde{\Omega}_p = (p^\diamond, p^\diamond + \mathcal{M})$ and $p^\diamond = \eta_{\max}^2 T$; $\mathcal{M}$ is a length such that $\mathcal{M} \leq \min\{\pi L - \eta_{\max}^2 T, 1/2\eta_{\max}^2 T\}$. There holds

$$(4.25) \quad \|u_0 - u_h^d\|_{L^2(\tilde{\Omega}_p) \times L^2(\Omega_x)} \lesssim (\Delta p + \Delta x^s) e^{\eta_{\max}^2 T + \mathcal{M}} \|w_0\|_{H^s(\Omega_x) \times H^1(\Omega_p)} + \varepsilon.$$

Proof. From the recovery rule in Theorem 3.2, it follows that

$$(4.26) \quad \|u_0 - u_h^d\|_{L^2(\tilde{\Omega}_p) \times L^2(\Omega_x)} = \|e^p w_0 - e^p w_h^d\|_{L^2(\tilde{\Omega}_p) \times L^2(\Omega_x)}.$$

Using the triangle equality and (4.16), it yields

$$\begin{aligned} (4.27) \quad &\|u_0 - u_h^d\|_{L^2(\tilde{\Omega}_p) \times L^2(\Omega_x)} \\ &\leq \|e^p \mathcal{W}_0 - e^p w_h^d\|_{L^2(\tilde{\Omega}_p) \times L^2(\Omega_x)} + \|e^p w_0 - e^p \mathcal{W}_0\|_{L^2(\tilde{\Omega}_p) \times L^2(\Omega_x)} \\ &\leq (\Delta p + \Delta x^s) e^{\eta_{\max}^2 T + \mathcal{M}} \|w_0\|_{H^s(\Omega_x) \times H^1(\Omega_p)} + e^{\eta_{\max}^2 T + \mathcal{M}} \|w_0 - \mathcal{W}_0\|_{L^2(\Omega_x) \times L^2(\tilde{\Omega}_p)}. \end{aligned}$$

The proof is completed by using Lemma 4.2, (4.4), and $\mathcal{M} \leq \frac{1}{2}\eta_{\max}^2 T$. $\square$

As can be seen from Theorem 3.2, the error increases as the computation time progresses from T to 0, which is also reflected in the numerical calculations. The error we provided here corresponds to the error at the moment $t = 0$, which is the worst-case scenario. This aligns with the numerical test in subsection 6.2, where the error increases as the time approaches zero. Consequently, the error estimation in Theorem 4.3 can be considered as taking the L^∞ norm in the time direction.

We point out that our algorithm offers three significant advantages. Firstly, the Hamiltonian nature of the semidiscrete system (4.5) allows for the development of a

computationally stable scheme, provided the CFL condition is met. Secondly, Theorem 3.2 ensures a straightforward recovery of the original variable $\mathbf{u}_h$ from $\mathbf{w}_h$. Although a time-dependent exponential growth factor $e^{\eta_{\max}^2 T}$ is introduced at this step, we highlight that the value of $\eta_{\max}$ is relatively small, as noted in Remark 3.4. Furthermore, by choosing an appropriate interval in the extended space to recover the original variables, essentially truncating the Fourier modes of the input data, the originally ill-posed problem becomes well-posed. Although the initial-value problem to the Schrödingerized equation is well-posed even without this Fourier truncation, to recover $\mathbf{u}_h$ one needs *finite* Fourier mode. Thirdly, our algorithm provides a quantum algorithm for solving ill-posed problems that can be implemented on a quantum computer (see section 7).

4.3. A higher-order improvement in p . The first-order convergence in p was due to the lack of regularity in w , which is continuous but not in C^1 . The convergence rate can be improved by choosing a smoother initial data for w in the extended space. For example, consider the following setup:

$$(4.28) \quad \frac{d}{dt}w = -\partial_{xxp}w \quad \text{in } \Omega_x \times \Omega_p, \quad T > t \geq 0, \quad w(T) = g(p)u_T(x) \quad \text{in } \Omega_x \times \Omega_p,$$

where $g(p)$ defined in $\mathbb{R}$ satisfies

$$(4.29) \quad g(p) = \begin{cases} h(p), & p \in (-\infty, 0], \\ e^{-p}, & p \in (0, \infty), \end{cases}$$

with $h(p) \in L^2((-\infty, 0))$. It is easy to check the Fourier transform of w denoted by $\hat{w}$ satisfies

$$\frac{d}{dt}\hat{w} = -i\xi\eta^2\hat{w}, \quad \hat{w}(T, \xi, \eta) = \hat{g}(\xi)\hat{u}_T(\eta),$$

where $\hat{g}(\xi) = \mathcal{F}(g)$, and $w(t, x, p) = \int_{\mathbb{R}} g(p + \eta^2(t - T))\hat{u}_T(\eta)e^{-ix\eta} d\eta$. In addition, the results in sections 2 and 3 still hold, since we do not care about the solution when $p < 0$.

From (4.16) and Theorem 4.3, the limitation of the convergence order mainly comes from the nonsmoothness of $g(p)$. In order to improve the whole convergence rates, we could smooth $g(p)$ by using higher-order polynomials, for example,

$$(4.30) \quad h(p) = \mathcal{P}_{2k+1}(p), \quad p \in [-1, 0], \quad h(p) = e^p, \quad p \in (-\infty, -1),$$

where $\mathcal{P}_{2k+1}$ is a Hermite interpolation [32, section 2.1.5] and satisfies

$$\begin{aligned} (\partial_{p^\alpha} \mathcal{P}_{2k+1}(p))|_{p=0} &= (\partial_{p^\alpha} (e^{-p}))|_{p=0} = (-1)^\alpha, \\ (\partial_{p^\alpha} \mathcal{P}_{2k+1}(p))|_{p=-1} &= (\partial_{p^\alpha} (e^p))|_{p=-1} = e^{-1}. \end{aligned}$$

Here $0 \leq \alpha \leq k$ is an integer. Therefore, $g \in C^k(\mathbb{R})$, and one has the estimate of u_h^d as

$$\|u_0 - u_h^d\|_{L^2(\hat{\Omega}_p) \times L^2(\Omega_x)} \lesssim (\Delta p^{k+1} + \Delta x^s) \|u_0\|_{H^s(\Omega_x)} + \varepsilon.$$

The choice of k can be chosen such that $\Delta p^{k+1} = \mathcal{O}(\Delta x^s)$. We use $k = 1$ in our numerical experiments in section 6.

The linear combination of Hamiltonian simulation (LCHS) method introduced in [2] employs the continuous Fourier transform in p and then applies numerical integration to complete the inverse of Fourier transform, choosing $p = 0$ to recover the

target variables u . Initially, it is a first-order method. Subsequent enhancements in [1] improve the accuracy of the original LCHS algorithms by introducing new kernel functions with faster decay and truncating the Fourier transform integral to a finite interval $[-X, X]$, where $X = (\log(1/\varepsilon))^{1/\beta}$ ($0 < \beta < 1$) truncates the continuous Fourier variable ξ . We emphasize that the fundamental principle remains consistent with our approach to smooth extension. Indeed, the Fourier transforms of the kernel functions are e^{-p} for $p > 0$, whereas they exhibit significant differences on the negative real axis. Therefore, the transformed kernel function can be interpreted as a smooth extension in the p space.

5. Application to the convection equations with purely imaginary wave speed. In this section, we concentrate on the linear convection equation with an imaginary convection speed which is unstable and thus hard to simulate numerically. This is a simple model for more complicated, physically interesting problems such as Maxwell's equation with negative index of refraction [30, 28] that has applications in meta materials. We consider the one-dimensional model with periodic boundary condition

$$(5.1) \quad \partial_t v = i \partial_x v \quad \text{in } \Omega_x, \quad v(x, 0) = v_0,$$

where $\Omega_x = [-\pi, \pi]$. This problem is ill-posed since, by taking a Fourier transform on x , one gets

$$\partial_t \hat{v} = -\eta \hat{v},$$

where $\hat{v}$ is the Fourier transform of v in x . Assuming the solution of (5.1) in Sobolev space exists, the exact solution is

$$(5.2) \quad v(x, t) = \int_{\mathbb{R}} e^{-ix\eta} e^{-\eta t} \hat{v}_0 \, d\eta.$$

Clearly, the solution contains exponential growing modes corresponding to $\eta < 0$. Suppose $\hat{v}_0 \in H_0^s$; then $v(t, x) \in H^s(\Omega_x)$. The Schrödingerization gives

$$(5.3) \quad \frac{d}{dt} w = -i \partial_{xp} w \quad \text{in } \Omega_x \times \Omega_p, \quad w(0, x, p) = e^{-|p|} v_0(x) \quad \text{in } \Omega_x \times \Omega_p,$$

where $\Omega_p = (-\infty, \infty)$. The Fourier transform shows

$$(5.4) \quad \frac{d}{dt} \hat{w} = i\xi\eta\hat{w}, \quad \hat{w}(0, \xi, \eta) = \frac{\hat{v}_0(\eta)}{\pi(1 + \xi^2)},$$

which has a bounded solution for all time, with the exact solution given by

$$w(t, x, p) = \int_{\mathbb{R}} \int_{\mathbb{R}} e^{-i(x\eta + p\xi)} e^{i\xi\eta t} \hat{w}_0 \, d\eta d\xi = \int_{\mathbb{R}} e^{-|p - \eta t|} \hat{v}_0(\eta) e^{-ix\eta} \, d\eta.$$

Obviously, the w -equation (5.3) and the Hamiltonian system (5.4) can be simulated by a stable scheme in both quantum computers and classical ones. We define an approximate solution to the truncation as follows.

DEFINITION 5.1. *Define*

$$(5.5) \quad w_{\eta_{\max}} = \int_{\mathbb{R}} e^{-|p - \eta t|} \hat{v}_0 \chi_{\max} e^{-ix\eta} \, d\eta,$$

where $\chi_{\max}$ is a characteristic function of $[-\eta_{\max}, \eta_{\max}]$. The truncated solution is

$$(5.6) \quad v_{\eta_{\max}}(t, x, p) = e^p w_{\eta_{\max}}(t, x, p), \quad \text{or} \quad v_{\eta_{\max}}(t, x, p) = e^p \int_p^\infty w(t, x, q) \, dq, \quad p > p^\diamond,$$

with $p^\diamond = \eta_{\max} T$.

Following Theorem 3.2, one gets the estimate for recovery. The proof is omitted here.

THEOREM 5.2. *Let $v(t, x)$ and $w(t, x, p)$ be the exact solution of (5.1), (5.3), respectively, on the interval $t \in [0, T]$. Suppose there exists an a priori bound $\|v_T\|_{H^s(\Omega_x)}$ and $v_{\eta_{\max}, T} = v_{\eta_{\max}}(T, x, p)$ is given by (5.6), where $p > p^\diamond = \eta_{\max} T$. Then*

$$(5.7) \quad \|v_T - v_{\eta_{\max}, T}\|_{L^2(\Omega_x)} \leq \frac{\delta(\eta_{\max})}{\eta_{\max}^s},$$

where $\delta(\eta_{\max}) = (\int_{\eta_{\max}}^\infty (1 + \eta^2)^s |\hat{v}_T|^2 \, d\eta)^{\frac{1}{2}}$.

Similar to Remark 3.4, it follows from $v_T \in H^s(\Omega_x)$ that $\delta(\eta_{\max})$ is a small number. In order to keep the error within ε , one could choose $\eta_{\max} \geq (\frac{\delta}{\varepsilon})^{\frac{1}{s}}$. For example, if $u = \alpha(t) \cos(\omega_0 x)$, then one gets $v_T = e^p w(T, x, p)$ for $p > p^\diamond = \eta_{\max} T = \omega_0 T$.

5.1. Discretization for Schrödingerization of the convection equations with imaginary convective terms. In this section, we give the discretization of Schrödingerization for the convection equations with purely imaginary wave speed. If we use the spectral method, the algorithm is quite similar to subsection 4.1. It is obtained by letting $A = D_x$ in (4.5). It is obvious to see that A is Hermitian and has both positive and negative eigenvalues. It is hard to construct a stable scheme if one solves the original problem. However, after Schrödingerization, it becomes a Hamiltonian system which is quite easy to approximate in both classical and quantum computers. Applying Theorem 5.2, one gets the desired variables.

6. Numerical tests. In this section, we perform several numerical simulations for the backward heat equation and imaginary convection equation by Schrödingerization in order to check the performance (in recovery and convergence rates) of our methods. For all numerical simulations, we perform the tests in the classical computers by using the Crank–Nicolson method for temporal discretization. In addition, we apply the trapezoidal rule for numerical integration of all of the tests. In order to get higher-order convergence rates of discretization (4.5), we smooth the initial data $w(T)$ in (3.5) by choosing

$$(6.1) \quad g(p) = \begin{cases} (-3 + 3e^{-1})p^3 + (-5 + 4e^{-1})p^2 - p + 1, & p \in (-1, 0), \\ e^{-|p|}, & p \in (-\infty, -1] \cup [0, \infty). \end{cases}$$

Therefore, $g(p) \in H^2(\Omega_p)$.

6.1. Approximation of smooth solutions. These tests are used to evaluate the recovery and the order of accuracy of our method. More precisely, we want to show the choice of $p^\diamond$ in (3.18) and convergence rates in Theorems 4.3.

In the first test, we consider the backward heat equation in $\Omega_x = [0, 2]$ with Dirichlet boundary condition $u(T, 0) = u(T, 2) = 0$ and $T = 1$. We take a smooth solution

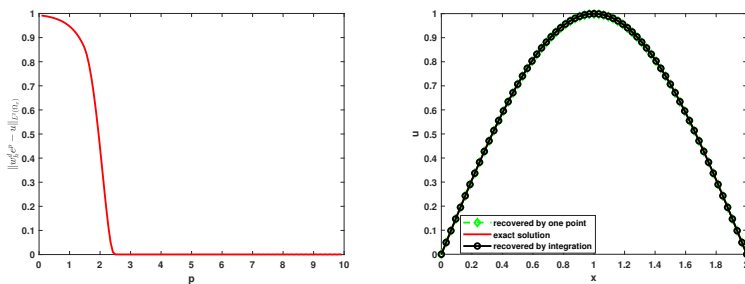


FIG. 1. Left: $\|w_h^d e^p - u\|_{L^2(\Omega_x)}$ with respect to p , with w_h^d computed by (4.5)–(4.10). Right: the recovery from Schrödingerization by choosing $p > p^\diamond = \pi^2/4$.

TABLE 1

The convergence rates of $\|u_h^d - u\|_{L^2(\Omega_x) \times L^2(\tilde{\Omega}_p)}$ and $\|u_d^* - u\|_{L^2(\Omega_x)}$, where u_h^d is computed by (4.5)–(4.11) and $u_d^* = \frac{\int_3^{10} w_h^d dp}{e^{-3} - e^{-10}}$, $\tilde{\Omega}_p = [3, 10]$ and $\Delta t = \frac{\Delta x}{2^3}$.

$(\Delta p, \Delta x)$	$(\frac{10}{2^7}, \frac{1}{2^5})$	order	$(\frac{10}{2^8}, \frac{1}{2^6})$	order	$(\frac{10}{2^9}, \frac{1}{2^7})$	order
$\ u_h^d - u\ _{L^2(\Omega_x) \times L^2(\tilde{\Omega}_p)}$	1.64e-03	-	3.25e-04	2.33	8.05e-05	2.01
$\ u_d^* - u\ _{L^2(\Omega_x)}$	7.48e-04	-	1.86e-04	2.01	4.56e-05	2.00

$$(6.2) \quad u = \exp\left(-\frac{\pi^2}{4}t\right) \sin\left(\frac{\pi x}{2}\right),$$

and the initial data is $u_T = \exp(-\frac{\pi^2}{4}) \sin(\frac{\pi x}{2})$. We use the finite difference method for spatial discretization with A defined by

$$(6.3) \quad A = \frac{1}{\Delta x^2} \begin{bmatrix} -2 & 1 & & & \\ 1 & -2 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & 1 & -2 & 1 \\ & & & 1 & -2 \end{bmatrix},$$

where Δx is the mesh size. The computation stops at $t=0$. According to (3.18), one has $\eta_{\max} = \frac{\pi}{2}$ and $p^\diamond = \frac{\pi^2}{4}$. The numerical results are shown in Figure 1 and Table 1. From the plot on the left in Figure 1, it can be seen that the error between u and $u_h^d = w_h^d e^p$ drops precipitously at $p^\diamond$, and numerical solutions of Schrödingerization recovered by choosing one point or numerical integration are close to the exact solution. According to Table 1, second-order convergence rates of $\|w_h^d e^p - u\|_{L^2(\Omega_x) \times L^2(\tilde{\Omega}_p)}$ and $\|u_d^* - u\|_{L^2(\Omega_x)}$ are obtained, respectively, where $u_d^* = \frac{\int_3^{10} w_h^d dp}{e^{-3} - e^{-10}}$.

Next, we consider another exact solution set by

$$(6.4) \quad u = \exp(-9\pi^2 t) \sin(3\pi x) \quad \text{in } [0, 2],$$

with periodic boundary conditions and the initial data is $u_T = \exp(-9\pi^2) \sin(3\pi x)$. We apply the spectral method to discrete the x -domain. From Theorem 3.2 and Remark 3.4, we deduce that $p^\diamond = \omega_0^2 T = 9\pi^2 \approx 90$. From Figure 2, it can be seen that $u = e^p w$ when $p \geq p^\diamond \approx 90$, and the numerical solutions are very close to the exact solution. In Table 2, the L^2 errors of the method (4.5)–(4.11) are presented.

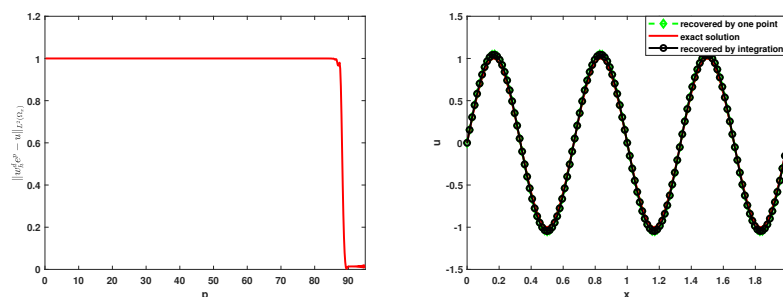


FIG. 2. Left: $\|w_h^d e^p - u\|_{L^2(\Omega_x)}$ with respect to p , with w_h^d computed by (4.5)–(4.10). Right: the recovery from Schrödingerization by choosing $p > p^\diamond = 9\pi^2$.

TABLE 2

The convergence rates of $\|u_h^d - u\|_{L^2(\Omega_x) \times L^2(\tilde{\Omega}_p)}$ and $\|u_d^* - u\|_{L^2(\Omega_x)}$, respectively, where u_h^d is computed by (4.10)–(4.11) and $u_d^* = \frac{\int_{90}^{95} w_h^d dp}{e^{-90} - e^{-95}}$, $\tilde{\Omega}_p = [90, 95]$, $\Delta t = \frac{\Delta x}{2^4}$.

$(\Delta p, \Delta x)$	$(\frac{95}{2^8}, \frac{1}{2^6})$	order	$(\frac{95}{2^9}, \frac{1}{2^7})$	order	$(\frac{95}{2^{10}}, \frac{1}{2^8})$	order
$\ u_h^d - u\ _{L^2(\Omega_x) \times L^2(\tilde{\Omega}_p)}$	4.73e-01	-	1.21e-01	1.96	3.07e-02	1.97
$\ u_d^* - u\ _{L^2(\Omega_x)}$	1.37e-01	-	5.08e-02	1.43	1.35e-02	1.90

They show the optimal orders $\|u_h^d - u\|_{L^2(\Omega_x) \times L^2(\tilde{\Omega}_p)} \sim \Delta p^2$, $\|u_d^* - u\|_{L^2(\Omega_x)} \sim \Delta p^2$ are obtained when $\Delta p \rightarrow 0$, where $\tilde{\Omega}_p = [90, 95]$, and $u_h^d = e^p w_h^d$, $u_d^* = \frac{\int_{90}^{95} w_h^d dp}{e^{-90} - e^{-95}}$.

6.2. Approximation of a piecewise smooth solution. Now, we consider a piecewise smooth solution in $(-\infty, \infty)$,

$$(6.5) \quad u(x, 0) = \begin{cases} -|x| + 1, & x \in [-1, 1], \\ 0 & \text{otherwise.} \end{cases}$$

The tests are done in the interval $x \in [-20, 20]$ and $\Delta x = \frac{40}{2^7}$, $\Delta t = \frac{1}{2^8}$. We use the finite difference method to simulate the heat equation with the initial condition given by (6.5). Thus, we get the approximate solution to u_T at $T = 1$. According to Theorem 3.2, we choose $\varepsilon = \Delta x$, $\delta(\eta_{\max}) = 1$ and $s = 2, 2$, and 1 for recovering the target variables $u(t)|_{t=0.5}$, $u(t)|_{t=0.25}$, and $u(t)|_{t=0}$, respectively. Thus, it yields $\eta_{\max} = (\frac{1}{\Delta x})^{1/2} = \sqrt{3.2}$, $(\frac{1}{\Delta x})^{1/2} = \sqrt{3.2}$, and $\frac{1}{\Delta x} = 3.2$. The results are shown in Figure 3 with u_h^d computed by (4.5)–(4.11) and $p \in [-30, 30]$, $\Delta p = \frac{60}{2^{10}}$. From the plot of Figure 3, we find that the error is larger when t is smaller, which is consistent with the estimate in Theorem 3.2.

6.3. Input data with noises. In this test, we investigate the cases with the input data containing noises. The exact solution in $\mathbb{R}$ is set by

$$(6.6) \quad u(x, t) = \frac{1}{\sqrt{1 + 4t}} e^{-\frac{x^2}{1 + 4t}}.$$

The input data is $u_T^\zeta = u_T + \zeta_0 \text{rand}(x)$. The magnitude ζ_0 indicates the noise level of the measurement data, and $\text{rand}(x)$ is a random number such that $-1 \leq \text{rand}(x) \leq 1$. The tests are truncated to the interval $[-10, 10]$ and $T = 1$. The noise levels are $\zeta_0 = 6 \times 10^{-2}$, 6×10^{-3} , and 6×10^{-4} . We apply (4.5)–(4.11) to discretize the

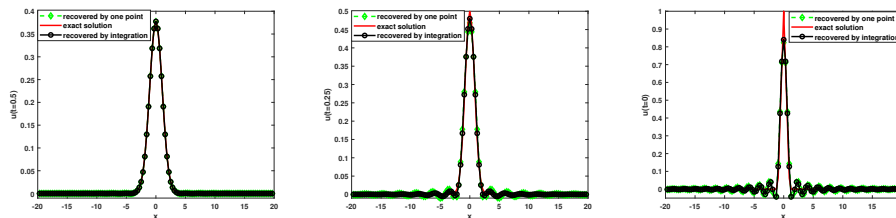


FIG. 3. The exact solution in the legend is the numerical solution at time $t = 0.5$, $t = 0.25$, and $t = 0$, respectively, obtained by solving the forward heat conduction equation using the finite difference method. The results of Schrödingerization are shown with $\eta_{\max} = (\frac{1}{\Delta x})^{1/2} = \sqrt{3.2}$, $(\frac{1}{\Delta x})^{1/2} = \sqrt{3.2}$, $\frac{1}{\Delta x} = 3.2$, respectively.

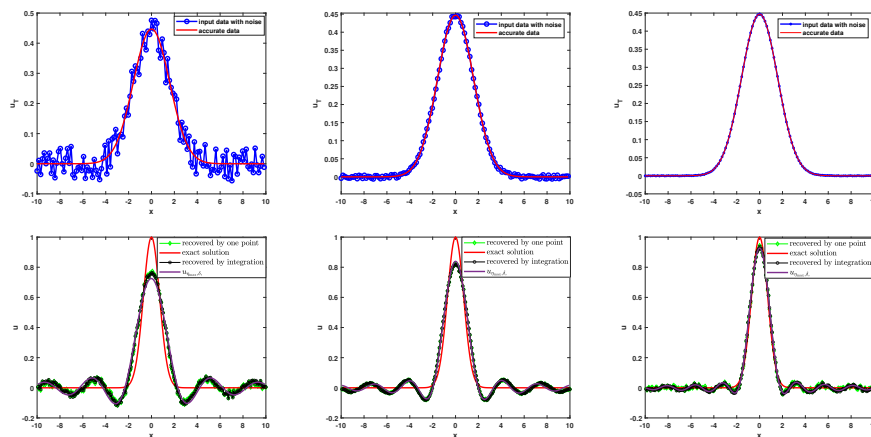


FIG. 4. The first row: input data with different noise levels from left to right. The noise levels are 6×10^{-2} , 6×10^{-3} , and 6×10^{-4} . The second row: results of Schrödingerization computed by (4.5)–(4.11) using the input data above with $\eta_{\max} = 1.5, 2, 2.5$, respectively.

Schrödingerization with $p \in [-20, 20]$ and $\Delta p = \frac{10}{2^6}$. We use the finite difference method to discretize the x -domain with $\Delta x = \frac{10}{2^6}$, $\Delta t = \frac{1}{2^{10}}$. The results are shown in Figure 4. According to Theorem 3.5, it is hard to find $\eta_{\max}$ to satisfy $\frac{\delta(\eta_{\max})}{\eta_{\max}} + e^{\eta_{\max}^2 T} \zeta_0 \leq \varepsilon$ for any fixed ζ_0 , T and ε . However, one could choose $\eta_{\max} = \sqrt{\ln((\frac{1}{\zeta_0})^{\frac{1}{T}} (\ln \frac{1}{\zeta_0})^{\frac{-s}{2T}})}$ to get $\|u(t) - u_{\eta_{\max}, \zeta}(t)\|_{L^2(\Omega_x)} \lesssim (\ln(\frac{1}{\zeta_0}))^{-\frac{s}{2T}} \rightarrow 0$ as ζ_0 tends to zero under the assumption $T \leq \ln(1/\zeta_0)$, $\zeta_0 < 1$. By investigating the cases of different ζ_0 , we find the numerical solution of Schrödingerization approximates $u_{\eta_{\max}, \zeta}$ and tends to the exact solution as ζ_0 goes to zero. The oscillation of the numerical solution comes from two aspects: one is the truncation of the calculation area of x , and the other is the finite difference method of calculating the discontinuous input data.

6.4. The convection equation with imaginary wave speed. Now we consider the convection equations with imaginary wave speed to test the accuracy of the recovery. In this test, the exact solution is

$$(6.7) \quad v(t, x) = \cos(3\pi(x + t)),$$

and the simulation stops at $T = 1$. Therefore the source term is obtained by $f = \partial_t u - i\partial_x u$ with u defined in (6.7). We use the spectral method on p and x , then get the following linear system:

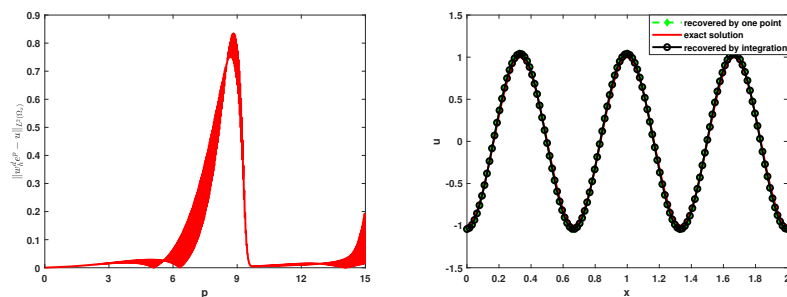


FIG. 5. Left: $\|w_h^d e^p - v\|_{L^2(\Omega_x)}$ with respect to p , where w_h^d is from (4.5). Right: the recovery from Schrödingerization by choosing $p > p^\diamond = 10$, where the numerical solution is obtained by $v_d^* = \frac{\int_{-10}^{15} w_h^d dp}{e^{-10} - e^{-15}}$, $p \in [-20, 20]$, $\Delta p = \frac{10}{2^8}$, and $\Delta x = \frac{1}{2^6}$.

$$(6.8) \quad \frac{d}{dt} \tilde{w}_h = i(D_p \otimes D_x) \tilde{w}_h + \tilde{f}_h, \quad \tilde{w}_h(0) = (\Phi_p^{-1} \otimes \Phi^{-1}) w_h(0),$$

where $\tilde{f}_h = (\Phi_p^{-1} \otimes \Phi^{-1}) f_h$, with $f_h = \sum_{j=0}^{M-1} \sum_{l=0}^{N-1} f(x_j, t) e^{-|p_l|} (e_l^{(N)} \otimes e_j^{(M)})$. In order to obtain unitary dynamical systems to facilitate operation on quantum computers, one needs to use the technique of dealing with source terms in [15]. In Figure 5, one can find $p^\diamond \approx 3\pi T$, which confirms Theorem 5.2, and the numerical solutions are close to the exact one. Since the solution is not smooth enough when using the spectral method, the oscillation appears in the error between v and v_h^d in Figure 5.

7. Quantum algorithms for ill-posed problems. The above numerical scheme is a new scheme that can be used on both classical and quantum devices based on qubits or qumodes (continuous variables) [14].

7.1. Qubit-based quantum computing. Since (4.5) is a Hamiltonian system, a quantum simulation can be carried out such as quantum signal processing [24], linear combination of unitaries [7], etc. The complete circuit for implementing the quantum simulation of $|\mathbf{u}_h(t)\rangle = \mathbf{u}_h(t)/\|\mathbf{u}_h(t)\|$ is illustrated in Figure 6, where $|\mathbf{u}_h(t)\rangle$ is a quantum state.

For the specific ill-posed problems, the differences from well-posed problems lie in (i) the cost of unitarily evolving the system in (4.5), and (ii) the cost of final measurement to retrieve $|\mathbf{u}_h(t)\rangle \equiv \mathbf{u}_h(t)/\|\mathbf{u}_h(t)\|$, $t < T$, depending on the specification of possible $p^\diamond$ for different problems.

Since the evolution is unitary, the cost analysis is similar whether one is running forward or backward in time. The only difference is that the initial state used in the forward equation is $|\mathbf{u}_h(0)\rangle$ plus the ancilla state, whereas in the backward equation it is $|\mathbf{u}_h(T)\rangle$ plus the same ancilla state (see Figure 6). In quantum algorithms, Hamiltonian simulation with nearly optimal dependence on all parameters can be found in [5], with complexity given by the next lemma.

LEMMA 7.1. *For a Hamiltonian system, $\frac{d}{dt} \mathbf{u} = -iH\mathbf{u}$ with H a Hermitian matrix acting on m_H qubits can be simulated within error ε with $\mathcal{O}(\tau \log(\frac{\tau}{\varepsilon})/\log \log(\frac{\tau}{\varepsilon}))$ queries and $\mathcal{O}(\tau m_H L_{\text{polylog}})$ additional 2-qubits gates, where $\tau = s(H)\|H\|_{\max} T$, $s(H)$ is the sparsity of H (maximum number of nonzero entries in each row), $\|H\|_{\max}$ is the max-norm (value of largest entry in absolute value), T is the evolution time, and*

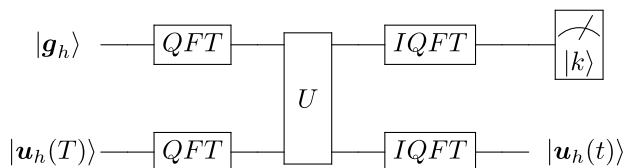


FIG. 6. Quantum circuit for Schrödingerization of (4.2), U is a unitary matrix approximating $U = \exp(i(D_p \otimes A)(T - t))$, and QFT (IQFT) denotes the quantum Fourier transform (inverse quantum Fourier transform), which is the quantum counterpart of the classical Fourier transform.

$$L_{polylog} = \left[1 + \log_{2.5}(\tau/\varepsilon)\right] \frac{\log(\tau/\varepsilon)}{\log \log(\tau/\varepsilon)}.$$

In the case of (4.5), since the spatial discretization is a second-order scheme $\Delta x^2 \sim \varepsilon$, we need $\mathcal{O}(d \log(\frac{1}{\sqrt{\varepsilon}}))$ qubits for x variable where d is the dimension. It is crucial to note that if one employs the smooth initialization method for the Schrödingerized equation, as introduced in subsection 4.3, and discretizes this smooth function in terms of p instead, both the initial data and the solution in p will belong to the class C^k for any integer $k \geq 1$. Consequently, if a spectral method is utilized, this approach will yield essentially spectral accuracy in the approximations within the extended p space. The extra qubits used in the extension domain p is almost $\mathcal{O}(\log \log(\frac{1}{\varepsilon}))$. Therefore, the required number of qubits is

$$m_H = \mathcal{O}\left(d \log\left(\frac{1}{\sqrt{\varepsilon}}\right) + \log \log\left(\frac{1}{\varepsilon}\right)\right),$$

the Hermitian matrix $H = D_p \otimes A$ satisfies $\|H\|_{\max} = \mathcal{O}(\frac{1}{\varepsilon} \log(\frac{1}{\varepsilon}))$, and the sparsity $s(H) = \mathcal{O}(d)$. By applying Lemma 7.1, one can directly obtain the complexity of Schrödingerization as follows.

LEMMA 7.2. *Given sparse-access to the Hermitian matrix, $H = D_p \otimes A$, and the initial quantum state $|u_h(T)\rangle$ has been obtained. The quantum state $|u_h(t)\rangle$ can be prepared to precision ε with $\tilde{\mathcal{O}}(\frac{dT}{\varepsilon})$ queries and $\tilde{\mathcal{O}}(\frac{d^2T}{\varepsilon})$ additional 2-qubits gates, where $\tilde{\mathcal{O}}$ denotes the case where all logarithmic factors are suppressed.*

This reduces the complexity of the Schrödingerization-based quantum algorithms for any dynamical system introduced in [16, 17], wherein the number of 2-qubit gates required for the heat equation is $\tilde{\mathcal{O}}(\frac{d^2T}{\varepsilon^2})$. Furthermore, this quantum approach demonstrates polynomial-level acceleration compared to classical algorithms for solving the forward heat equation.

Different values of $p^\diamond$ affect the probability of retrieving the correct final quantum state $|u_h(t)\rangle$ for $t < T$. The final step involves measurement of p . There are in principle two schemes from Theorem 2.1, where both schemes give a success probability proportional to $K(\|u(t)\|_{L^2(\Omega_x)}^2 / \|u(T)\|_{L^2(\Omega_x)}^2)$, and thus $\mathcal{O}(K^{-1} \|u(T)\|_{L^2(\Omega_x)}^2 / \|u(t)\|_{L^2(\Omega_x)}^2)$ measurements are needed, where K is a factor depending on $p^\diamond = \eta_{\max}^2 T$; for instance, see [16, 14]. The first scheme is the recovery by measurement of one point in (2.4). This corresponds to accepting the resulting state so long as the measured $p > \eta_{\max}^2 T$. In the qubit context for example, the largest value $K \sim e^{-2\eta_{\max}^2 T} \sim e^{-2(\frac{\varepsilon}{2})^{\frac{2}{s}} T}$, if $u_T \in H^s(\Omega_x)$. The second scheme is the recovery through the integration method in (2.5). For a bounded domain, the retrieval process then involves the projection

of $|w(T)\rangle$ onto the $[\eta_{\max}^2 T, \pi L]$ domain. In this case, $K \sim (\int_{p^\diamond}^{\pi L} e^{-p} dp)^2 = (e^{-\pi L} + e^{-p^\diamond})^2 \leq (e^{-\pi L} + e^{-\eta_{\max}^2 T})^2 \sim (e^{-\pi L} + e^{-(\frac{\delta}{\varepsilon})^{\frac{2}{\delta}}})^2$. Thus, for large domains where $\exp(-\pi L) \sim 0$, the first and second schemes are comparable. However, for specific cases, such as $u = \alpha(t) \cos(\omega_0 x)$ in Remark 3.4, the probability to retrieving the state $|u_h(0)\rangle$ is about

$$\frac{e^{-2\eta_{\max}^2 T} \|u(0)\|_{L^2(\Omega_x)}^2}{\|u(T)\|_{L^2(\Omega_x)}^2} = \frac{e^{-2\eta_{\max}^2 T} \alpha(0)^2}{\alpha(T)^2} = \frac{e^{-2\eta_{\max}^2 T} e^{2\omega_0^2 T} \alpha(T)^2}{\alpha(T)^2} = e^{2(\omega_0^2 - \eta_{\max}^2)T},$$

which is close to 1 if we choose $\eta_{\max} = \omega_0$. This condition aligns with the requirement for choosing $\eta_{\max}$ to successfully recover u , as analyzed in Remark 3.4.

7.2. Qumode-based analog quantum computing. Equation (2.1) is already of Schrödinger's form and can be rewritten with $|w(t)\rangle = \int \mathbf{w}(t, p) |p\rangle dp / \|\mathbf{w}(t, p)\|$ obeying

$$(7.1) \quad \frac{\partial |w(t)\rangle}{\partial t} = -i(H \otimes \hat{P})|w(t)\rangle, \quad |w_0\rangle = \int e^{-|p|} |p\rangle dp |u_0\rangle,$$

in the continuous-variable formalism, and $\hat{P} \leftrightarrow -i\partial/\partial p$ is the momentum operator conjugate to the position operator $\hat{X}_p$, which has p as eigenvalues and $[\hat{X}_p, \hat{P}] = iI$. Since $H \otimes \hat{P}$ is clearly Hermitian, then $|w(0)\rangle$ evolves by unitary evolution generated by the Hamiltonian $H \otimes \hat{P}$. We note that here, since we actually want to run the original PDE backward in time, the $|u_0\rangle$ state above corresponds in fact to the state $|u(t=T)\rangle$. If we want the state at some $t < T$, then the time $\mathcal{T}$ to run (7.1) is over $\mathcal{T} \in [0, T-t]$.

In the continuous-variable formulation, how $\eta_{\max} < \infty$ arises is more subtle. The finite $\eta_{\max}$ can come instead from a kind of energy constraint $E_{\max} \sim \eta_{\max}^2$ on the continuous-variable ancillary quantum mode, where $\eta_{\max}$ is acting like the maximum momentum for the mode (conjugate momentum to p). At the measurement step, here $p_{\max}^\diamond = \pi L$ is actually a maximum value of measurable p by the physical measurement apparatus. The detailed analysis here is more subtle and depends on the physical system studied, so we leave this to future work.

8. Conclusions. In this paper we present computationally stable numerical methods for two ill-posed problems: the (constant and variable coefficient) backward heat equation and the linear convection equation with imaginary wave speed. The main idea is to use Schrödingerization, which turns typical ill-posed problems into unitary dynamical systems in one higher dimension. The Schrödingerized system is a Hamiltonian system valid both forward and backward in time, and hence suitable for the design of a computationally stable numerical scheme. A key idea is given to recover the solution to the original problem from the Schrödingerized system using data from appropriate domain in the extended space. Some sharp error estimates between the approximate solution and exact solution are provided and also verified numerically. Moreover, a smooth initialization is introduced for the Schrödingerized system which leads to requiring an exponentially lower number of ancilla qubits for qubit-based quantum algorithms for any nonunitary dynamical system.

In the future this technique will be generalized to more realistic and physically interesting ill-posed problems.

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